Part 1: Bayesian Model

Session 1: Mostly motivation
Session 2: Mostly definitions
Session 3: Mostly computation and examples

Part 2: Bayesian Inference

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Session 6: Simulation based prediction of ỹ and h(ỹ)

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Session 7: Bayesian hierarchical model and model checking

Part 3

BAYESIAN HIERARCHICAL MODEL

3.1 – Shortcomings of the iid model

When $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ is i.i.d., the statistical model is:

$$M = \left\{ \prod_{i=1}^{n} p(\tilde{y}_i|\theta); \theta \in \Omega \right\}$$

and therefore

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|\theta \sim \prod_{i=1}^{n} p(\tilde{y}_i|\theta),$$

where θ is the same for all $\tilde{y}_i$

This assumes that the probability model $p(\tilde{y}_i|\theta^*)$ in M generating all observations is the same!

In the i.i.d. setting, we summarize n observations through a single parameter value, θ.

This is often too simplistic.

Let $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ represent the number of votes for a candidate across n electoral precincts.

If we use the independent binomial(n_i,θ) model:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|\theta \sim \prod_{i=1}^{n} \text{Binomial}(n_i,\theta).$$

we implicitly assume that the probability of voting for that candidate, θ, is identical in every precinct.

In reality, voter behavior varies across regions due to demographics, local issues, or socioeconomic factors.

We can move away from the i.i.d. assumption by modelling θ through covariates with a logistic model,

$$\theta_i(x_i) = \frac{e^{\beta_0+\beta_1 x_{1i}+\ldots+\beta_{p-1}x_{p-1i}}}{1+e^{\beta_0+\beta_1 x_{1i}+\ldots+\beta_{p-1}x_{p-1i}}}.$$

summarizing n observations through p parameter values.

In practice, covariates might not be available, relationships may be more complex, and there may be differences between precincts that cannot be explained by covariates.

3.2 – Bayesian hierarchical model

Hierarchical models assume that the $\tilde{y}=(\tilde{y}_1,\ldots, \tilde{y}_n)$ are conditionally independent but not identically distributed.

The parameter value is different for each $\tilde{y}_i$,

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} p(\tilde{y}_i|\theta_i)$$

All θ_i share the same distribution,

$$(\theta_1,\ldots,\theta_n)|\gamma \sim \prod_{i=1}^{n} \pi(\theta_i|\gamma)$$

where γ is unknown but known to belong to Ω_γ and:

$$\gamma \sim \psi(\gamma)$$

where ψ(γ) is a (fully known) prior distribution for γ.

A Bayesian hierarchical model is a special case of a Bayesian model, defined hierarchically.

In the non-Bayesian literature the corresponding statistical models are called multilevel or random effects models.

3.3 – Hierarchical binomial model

$\tilde{y}=(\tilde{y}_1,\ldots, \tilde{y}_n)$ is the number of abstainers in n=1447 electoral precincts and $\theta = (\theta_1,\ldots,\theta_n)$ is the probability of abstaining in these precincts.

$\tilde{Y}_i$ is abstencio_i and n_i is elec_i

	nommuni	CIU	PSC	PP	ICV	ERC	ciudadanos	altres	elec	votants	blancs	nuls	abstencio
Abrera	1	886	1497	561	439	476	139	149	7864	4234	63	24	3630
Aguilar de Segarra	1	71	14	9	12	19	0	5	200	137	5	2	63
Alella	1	1846	658	455	360	636	160	95	6760	4336	109	17	2424
Alpens	1	99	11	2	13	50	1	0	233	185	9	0	48
Ametlla del Valles, l	1	1400	530	286	279	560	101	55	5364	3312	90	11	2052
Arenys de Mar	1	1153	527	270	257	565	81	51	4595	2975	59	12	1620
Arenys de Mar	2	1296	850	307	304	592	91	96	6019	3616	63	17	2403
Arenys de Munt	1	772	342	112	161	478	28	38	3109	1996	61	4	1113
Arenys de Munt	2	684	303	88	150	388	19	30	2729	1693	27	4	1036
Argen\xe7ola	1	47	8	8	12	36	0	3	190	154	36	4	36
Argentona	1	1520	638	345	363	528	78	88	5976	3648	71	17	2328
Argentona	2	684	293	124	195	268	20	32	2655	1657	34	7	998
Art\xe9s	1	222	119	34	37	106	1	20	1011	561	17	5	450
Art\xe9s	2	659	368	101	127	465	8	37	2946	1818	38	15	1128
Avi\xe0	1	516	139	29	77	291	3	23	1686	1116	33	5	570
Aviny\xf3	1	745	109	26	51	267	4	5	1645	1237	25	5	408
Avinyonet del Penedes	1	318	158	30	71	133	4	14	1209	742	12	2	467
Aiguafreda	1	561	213	72	153	221	6	30	1879	1290	32	2	589
Badalona	1	2019	518	294	396	819	79	83	6216	4309	83	18	1907
Badalona	2	2376	1987	908	719	1065	290	201	14312	7717	144	27	6595

Example 1: Non-hierarchical binomial model

The non-hierarchical binomial model assumes that:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\theta \sim \prod_{i=1}^{n} \text{Binomial}(n_i, \theta).$$

and therefore that the probability of abstaining, θ, is identical in every precinct.

In reality, voter behavior varies across regions due to demographics, local issues, or socioeconomic factors.

Example 8: Hierarchical binomial model.

$\tilde{y}=(\tilde{y}_1,\ldots, \tilde{y}_n)$ is the number of abstainers in n=1447 electoral precincts, and $\theta = (\theta_1,\ldots,\theta_n)$ the probability of abstaining.

A hierarchical binomial model assumes:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|(\theta_1, \ldots, \theta_n) \sim \prod_{i=1}^{n} \text{Binomial}(n_i, \theta_i)$$

where the $\theta_i \in (0,1)$ share a distribution,

$$(\theta_1, \ldots, \theta_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{Beta}(\alpha, \beta)$$

where $(\alpha,\beta) \in (0,\infty)\text{x}(0,\infty)$ has prior distribution:

$$(\alpha, \beta) \sim \psi_1(\alpha)\psi_2(\beta) = \text{Exp}(a)\text{Exp}(b)$$

Example 8: Note that:

$$E(\theta_i|\alpha, \beta) = \alpha/(\alpha+\beta)$$

$$V(\theta_i|\alpha, \beta) = E(\theta_i|\alpha, \beta)(1-E(\theta_i|\alpha, \beta))/(1+\alpha+\beta)$$

Hence, given $E(\theta_i|\alpha,\beta)$ the larger α+β the smaller $\text{Var}(\theta_i|\alpha,\beta)$

When α+β is small, $\text{Var}(\theta_i|\alpha,\beta)$ is large, and precincts are highly diverse.

When α+β tends to infinity, $\text{Var}(\theta_i|\alpha,\beta)$ shrinks to zero, and we obtain the non-hierarchical binomial.

By estimating (α,β) from the data, we learn whether the precincts are similar enough to share information, or they should be treated as distinct entities.

Example 8: When we care about the distribution of θ_i, the statistical model for $\tilde{y}=(\tilde{y}_1,\ldots, \tilde{y}_n)$ can be posed as:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{BetaBinomial}(n_i, \alpha, \beta),$$

and the $\tilde{y}_i$ are i.i.d. BetaBinomial(n_i,α,β), with prior:

$$(\alpha, \beta) \sim \psi_1(\alpha)\psi_2(\beta) = ae^{-a\alpha}be^{-b\beta}$$

Example 8: Under this i.i.d. BetaBinomial(α,β),

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{BetaBinomial}(n_i, \alpha, \beta),$$

we have that:

$$E(\tilde{y}_i|\alpha, \beta) = n_i \frac{\alpha}{(\alpha+\beta)},$$

$$V(\tilde{y}_i|\alpha, \beta) = \frac{E(\tilde{y}_i|\alpha, \beta)(n_i - E(\tilde{y}_i|\alpha, \beta))}{n_i} \frac{n_i+\alpha+\beta}{1+\alpha+\beta}$$

Different from the binomial model, this one allows for $V(\tilde{y}_i|\alpha,\beta)$ to be larger than $E(\tilde{y}_i|\alpha, \beta)(n_i - E(\tilde{y}_i|\alpha, \beta))/n_i$., accomodating for possible overdispersion in the data.

Example 8: When we care about the distribution of $\tilde{y}_i$, the statistical model for $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ can be posed as:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} \text{Binomial}(n_i,\theta_i)$$

where the prior for $\theta = (\theta_1,\ldots,\theta_n) \in (0,1) \times \ldots \times (0,1)$ is:

$$(\theta_1,\ldots,\theta_n) \sim \int \prod_{i=1}^{n} \pi(\theta_i|\alpha,\beta)\psi_1(\alpha)\psi_2(\beta)d\alpha d\beta,$$

where $\pi(\theta_i|\alpha,\beta)$ is $\text{Beta}(\alpha,\beta)$.

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Example 8:

```
library(R2jags)

model.binhier <- "
model
{
  for (i in 1:nn) {
    y1[i] ~ dbin(pi1[i],Vot[i])
    pi1[i] ~ dbeta(alf1,bet1)
  }
  alf1 ~ dexp(1)
  bet1 ~ dexp(1)
  }
"

  data <- list(nn=length(P06$abstencio), y1=P06$abstencio, Vot=P06$elec)

  parameters <- c("pi1","alf1","bet1")

  example <-jags(data, parameters.to.save=parameters,
                model=textConnection(model.binhier), n.iter=3000, n.burnin=1000, n.thin=2,n.chains=3)

  print(example)
```

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Example 8:

```
>   print(example)
Inference for Bugs model at "3", fit using jags,
 3 chains, each with 3000 iterations (first 1000 discarded), n.thin = 2
 n.sims = 3000 iterations saved
```

	mu.vect	sd.vect	2.5%	25%	50%	75%	97.5%	Rhat	n.eff
alf1	11.164	0.431	10.338	10.879	11.154	11.450	12.053	1.008	950
bet1	17.684	0.681	16.377	17.228	17.673	18.111	19.095	1.007	1200
pi1[1]	0.461	0.006	0.450	0.458	0.461	0.465	0.473	1.001	3000
pi1[2]	0.324	0.031	0.265	0.303	0.324	0.344	0.384	1.001	3000
pi1[3]	0.359	0.006	0.347	0.355	0.359	0.363	0.370	1.001	3000
pi1[4]	0.227	0.025	0.179	0.209	0.226	0.243	0.279	1.004	620
pi1[5]	0.382	0.007	0.370	0.378	0.382	0.387	0.395	1.002	1800
pi1[6]	0.353	0.007	0.339	0.348	0.353	0.357	0.367	1.001	3000
pi1[7]	0.399	0.006	0.387	0.395	0.399	0.404	0.412	1.007	330
pi1[8]	0.358	0.008	0.342	0.352	0.358	0.364	0.375	1.001	3000
pi1[9]	0.380	0.009	0.362	0.374	0.380	0.386	0.398	1.003	950

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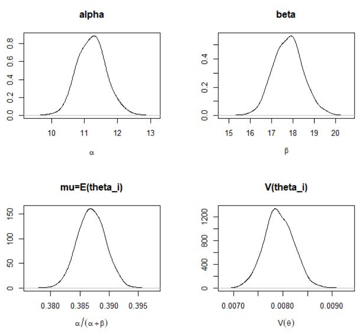
Example 8: Posterior distributions, abstention 2006:



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Example 8:

```
>   ### Estimation of 90% posterior credible interval
>   quantile(alf1,probs=c(.05,.5,.95))
       5%      50%      95% 
10.53514 11.22031 11.95832 
> 
>   ### Estimation of 90% posterior credible interval
>   quantile(bet1,probs=c(.05,.5,.95))
       5%      50%      95% 
16.68218 17.78287 18.91664 
```

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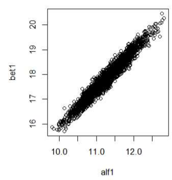
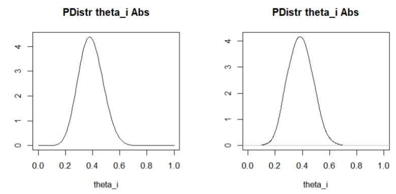
Example 8:





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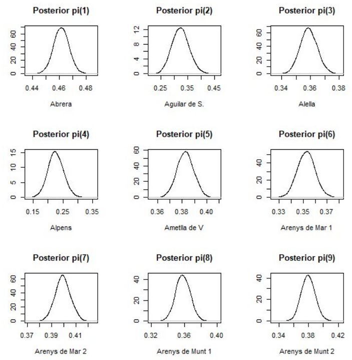
Example 8:



Example 8:

```
>   quantile(pi19,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3646033 0.3793672 0.3945553 
>   quantile(pi11,probs=c(.05,.5,.95))
       5%       50%       95% 
0.4521793 0.4614478 0.4709846 
>   quantile(pi12,probs=c(.05,.5,.95))
       5%       50%       95% 
0.2744154 0.3235282 0.3747721 
>   quantile(pi13,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3495517 0.3586236 0.3682133 
>   quantile(pi14,probs=c(.05,.5,.95))
       5%       50%       95% 
0.1848984 0.2251980 0.2697860 
>   quantile(pi15,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3719359 0.3826270 0.3932519 
>   quantile(pi16,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3411380 0.3528358 0.3645310 
>   quantile(pi17,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3888652 0.3990983 0.4091471 
>   quantile(pi18,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3442002 0.3579137 0.3726586 
>   quantile(pi19,probs=c(.05,.5,.95))
       5%       50%       95% 
0.3646033 0.3793672 0.3945553 
```


3.4 – Hierarchical Poisson model

Example 2: Let $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ be the number of goals scored, and θ the expected number of goals.

A Non-hierarchical model assumes:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|\theta \sim \prod_{i=1}^{n} \mathrm{Poisson}(\theta)$$

This requires θ to be the same for all games!

Goals scored by Girona FC in 2022/23 until April 11th: 0, 3, 0, 1, 2, 1, 3, 1, 1, 2, 1, 1, 2, 2, 2, 2, 2, 0, 0, 1, 0, 6, 3, 2, 0, 2, 2, 0.

Example 9: Hierarchical Poisson model.

Let $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ be the number of goals scored by GironaFC, and $\theta=(\theta_1,\ldots,\theta_n)$ their expected values. A hierarchical Poisson model assumes:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} \mathrm{Poisson}(\theta_i)$$

where all the $\theta_i \in (0,\infty)$ share a common distribution,

$$(\theta_1,\ldots,\theta_n)|\alpha,\beta \sim \prod_{i=1}^{n} \mathrm{Gamma}(\alpha,\beta)$$

where $(\alpha,\beta) \in (0,\infty)\mathrm{x}(0,\infty)$ has prior distribution:

$$(\alpha,\beta) \sim \psi_1(\alpha)\psi_2(\beta) = \mathrm{Exp}(a)\mathrm{Exp}(b)$$

Example 9: Note that

$$E(\theta_i|\alpha,\beta) = \alpha/\beta$$

$$V(\theta_i|\alpha,\beta) = E(\theta_i|\alpha,\beta)/\beta$$

Hence, given $E(\theta_i|\alpha,\beta)$ the larger β the smaller $Var(\theta_i|\alpha,\beta)$

When β tends to infinity $Var(\theta_i|\alpha,\beta)$ tends to zero and the limiting model is the non-hierarchical i.i.d. Poisson.

By learning about β one learns whether this hierarchical model is called for.

Example 9: When we care about the distribution of θ_i, the statistical model can be posed as:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{GammaPoisson}(\alpha, \beta),$$

and the observations are assumed to be i.i.d. GammaPoisson(α,β), with prior:

$$(\alpha, \beta) \sim \psi_1(\alpha)\psi_2(\beta) = ae^{-a\alpha}be^{-b\beta}$$

Example 9: When observations are i.i.d. GammaPoisson(α,β),

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{GammaPoisson}(\alpha, \beta),$$

We have that:

$$E(\tilde{y}_i|\alpha, \beta) = \frac{\alpha}{\beta},$$

$$V(\tilde{y}_i|\alpha, \beta) = E(\tilde{y}_i|\alpha, \beta)(1 + \frac{1}{\beta}).$$

Different from the i.i.d. Poisson(θ) model, this allows for $V(\tilde{y}_i|\alpha,\beta)$ to be larger than $E(\tilde{y}_i|\alpha,\beta)$ and not just equal to it.

Example 9: When we care about the distribution of $\tilde{y}_i$, the statistical model can be posed as:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|(\theta_1, \ldots, \theta_n) \sim \prod_{i=1}^{n} \text{Poisson}(\theta_i),$$

and the prior for $\theta = (\theta_1,\ldots,\theta_n) \in (0,\infty)\text{x}\ldots\text{x}(0,\infty)$ is:

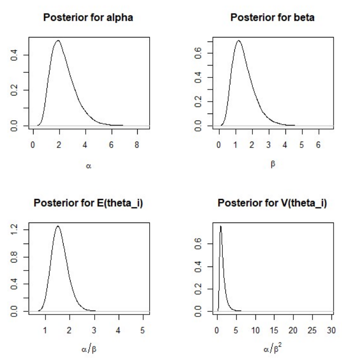
$$(\theta_1, \ldots, \theta_n) \sim \int \prod_{i=1}^{n} \pi(\theta_i|\alpha, \beta)\psi_1(\alpha)\psi_2(\beta) d\alpha d\beta,$$

where $\pi(\theta_i|\alpha, \beta)$ is $\text{Gamma}(\alpha, \beta)$.

Example 9:

```
library(R2jags)

model.poishier <- "
 model
 {
  for (i in 1:n) {
   y[i] ~ dpois(theta[i])
   theta[i] ~ dgamma(alf1,bet1)
   }
   alf1 ~ dexp(a)
   bet1 ~ dexp(b)
 }
"

data <- list(y=y, n=n, a=a, b=b)

parameters <- c("theta","alf1","bet1")

examplePoisshier <-jags(data, parameters.to.save=parameters,
                  model=textConnection(model.poishier),n.iter=55000,n.burnin=5000,n.thin=2,n.chains=3)

print(examplePoisshier)
```

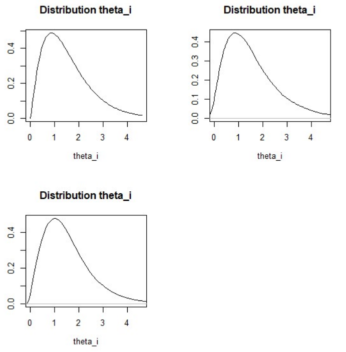

Example 9: Posterior distributions:



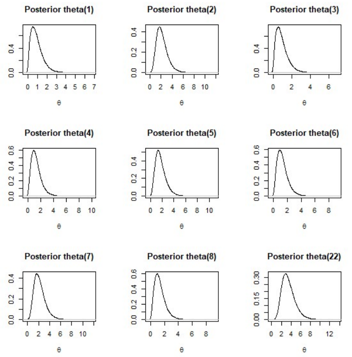
Example 9:

```
> ### Estimation of posterior expected value
> mean(alf1)
[1] 2.298683
> ### Estimation of posterior median
> median(alf1)
[1] 2.139889
> ### Estimation of 90% posterior credible interval
> quantile(alf1,probs=c(.05,.95))
      5%      95%
1.067326 4.056333
>
> ### Estimation of posterior expected value
> mean(bet1)
[1] 1.487106
> ### Estimation of posterior median
> median(bet1)
[1] 1.373717
> ### Estimation of 90% posterior credible interval
> quantile(bet1,probs=c(.05,.95))
       5%       95%
0.6413557 2.7121962
```

Example 9:



Example 9:



Example 9:

```
> mean(theta1)
[1] 0.9020328
> quantile(theta1,probs=c(.05,.95))
       5%       95%
0.1279068 2.1241050
> mean(theta2)
[1] 2.18117
> quantile(theta2,probs=c(.05,.95))
       5%       95%
0.8513542 4.0627039
> mean(theta3)
[1] 0.9021158
> quantile(theta3,probs=c(.05,.95))
       5%       95%
0.1299092 2.1296514
> mean(theta4)
[1] 1.330213
> quantile(theta4,probs=c(.05,.95))
       5%       95%
0.3453685 2.8118542
> mean(theta5)
[1] 1.756904
> quantile(theta5,probs=c(.05,.95))
       5%       95%
0.5967677 3.4494286
```

```
> mean(theta6)
[1] 1.334276
> quantile(theta6,probs=c(.05,.95))
       5%       95%
0.3542509 2.8084040
> mean(theta7)
[1] 2.189271
> quantile(theta7,probs=c(.05,.95))
       5%       95%
0.8500651 4.1018740
> mean(theta8)
[1] 1.333232
> quantile(theta8,probs=c(.05,.95))
      5%      95%
0.348574 2.811931
> mean(theta22)
[1] 3.470799
> quantile(theta22,probs=c(.05,.95))
      5%      95%
1.641032 6.008957
```


3.5 – Hierarchical exponential model

Example 10: Hierarchical exponential model.

Let $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ be the lifespan of black slugs and $\theta = (\theta_1,\ldots,\theta_n)$ the inverse of their expected values. The model assumes:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} \text{Exponential}(\theta_i)$$

where all the $\theta_i \in (0,\infty)$ share a common distribution,

$$(\theta_1,\ldots,\theta_n)|\alpha,\beta \sim \prod_{i=1}^{n} \text{Gamma}(\alpha,\beta)$$

where $(\alpha,\beta) \in (0,\infty) \times (0,\infty)$ has prior distribution:

$$(\alpha,\beta) \sim \psi_1(\alpha)\psi_2(\beta) = \text{Exp}(a)\text{Exp}(b)$$

Example 10: Note that:

$$E(\theta_i|\alpha,\beta) = \alpha/\beta$$

$$V(\theta_i|\alpha,\beta) = E(\theta_i|\alpha,\beta)/\beta$$

Hence, fixed $E(\theta_i|\alpha,\beta)$, the larger β the smaller $Var(\theta_i|\alpha,\beta)$

When β tends to infinity $Var(\theta_i|\alpha,\beta)$ tends to zero and the limiting model is the non-hierarchical i.i.d. exponential.

By learning about β one learns whether this hierarchical model is called for.

Example 10: When we care about the distribution of θ_i, the statistical model can be posed as:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{GammaExponential}(\alpha, \beta),$$

and the observations are assumed to be i.i.d. GammaExponential(α,β), with prior:

$$(\alpha, \beta) \sim \psi_1(\alpha)\psi_2(\beta) = a e^{-a\alpha} b e^{-b\beta}$$

Example 10: When observations i.i.d. GammaExponential(α,β),

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|\alpha, \beta \sim \prod_{i=1}^{n} \text{GammaExponential}(\alpha, \beta),$$

when α > 2:

$$V(\tilde{y}_i|\alpha, \beta) = E(\tilde{y}_i|\alpha, \beta)^2 \frac{\alpha}{\alpha - 2}$$

Different from the i.i.d. exponential(θ) model, this allows for $V(\tilde{y}_i|\alpha,\beta)$ to be larger than $E(\tilde{y}_i|\alpha,\beta)^2$ and not just equal to it, (but for α < 2 the variance becomes infinity).

Example 10: When we care about the distribution of $\tilde{y}_i$, the statistical model can be posed as:

$$(\tilde{y}_1, \ldots, \tilde{y}_n)|(\theta_1, \ldots, \theta_n) \sim \prod_{i=1}^{n} \text{Exponential}(\theta_i),$$

and the prior for θ = $(\theta_1,\ldots,\theta_n) \in (0,\infty) \times \ldots \times (0,\infty)$ is:

$$(\theta_1, \ldots, \theta_n) \sim \int \prod_{i=1}^{n} \pi(\theta_i|\alpha, \beta)\psi_1(\alpha)\psi_2(\beta) d\alpha d\beta,$$

where $\pi(\theta_i|\alpha, \beta)$ is $\text{Gamma}(\alpha, \beta)$.

Example 10:

```
library(R2jags)

model.exponhier <- "
 model
 {
  for (i in 1:n) {
   y[i] ~ dexp(theta[i])
   theta[i] ~ dgamma(alf1,bet1)
   }
   alf1 ~ dexp(a)
   bet1 ~ dexp(b)
 }
"

data <- list(y=y, n=n, a=a, b=b)

parameters <- c("theta","alf1","bet1")

exampleExponhier <-jags(data, parameters.to.save=parameters,
                  model=textConnection(model.exponhier),n.iter=55000,n.burnin=5000,n.thin=2,n.chains=3)

print(exampleExponhier)
```


3.6 – Hierarchical multinomial model

Example 11:

	nommuni	CIU	PSC	PP	ICV	ERC	ciudadanos	altres	elec	votants	blancs	nuls	abstencio
Abrera	1	886	1497	561	439	476	139	149	7864	4234	63	24	3630
Aguilar de Segarra	1	71	14	9	12	19	0	5	200	137	5	2	63
Alella	1	1846	658	455	360	636	160	95	6760	4336	109	17	2424
Alpens	1	99	11	2	13	50	1	0	233	185	9	0	48
Ametlla del Valles, l	1	1400	530	286	279	560	101	55	5364	3312	90	11	2052
Arenys de Mar	1	1153	527	270	257	565	81	51	4595	2975	59	12	1620
Arenys de Mar	2	1296	850	307	304	592	91	96	6019	3616	63	17	2403
Arenys de Munt	1	772	342	112	161	478	28	38	3109	1996	61	4	1113
Arenys de Munt	2	684	303	88	150	388	19	30	2729	1693	27	4	1036
Argen\xe7ola	1	47	8	8	12	36	0	3	190	154	36	4	36
Argentona	1	1520	638	345	363	528	78	88	5976	3648	71	17	2328
Argentona	2	684	293	124	195	268	20	32	2655	1657	34	7	998
Art\xe9s	1	222	119	34	37	106	1	20	1011	561	17	5	450
Art\xe9s	2	659	368	101	127	465	8	37	2946	1818	38	15	1128
Avi\xe0	1	516	139	29	77	291	3	23	1686	1116	33	5	570
Aviny\xf3	1	745	109	26	51	267	4	5	1645	1237	25	5	408
Avinyonet del Penedes	1	318	158	30	71	133	4	14	1209	742	12	2	467
Aiguafreda	1	561	213	72	153	221	6	30	1879	1290	32	2	589
Badalona	1	2019	518	294	396	819	79	83	6216	4309	83	18	1907
Badalona	2	2376	1987	908	719	1065	290	201	14312	7717	144	27	6595

Example 11: Hierarchical multinomial model.

Let $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ be the votes for k=10 candidates in n=1447 electoral precincts and $\theta = (\theta_1,\ldots,\theta_n)$ be the corresponding probabilities. A hierarchical multinomial model assumes:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} \text{Multinomial}(n_i,\theta_i),$$

where all the $\theta_i \in$Simplex of R^k are i.i.d. with,

$$(\theta_1,\ldots,\theta_n)|\tau,(\mu_1,\ldots,\mu_k) \sim \prod_{i=1}^{n} \text{Dirichlet}(\tau(\mu_1,\ldots,\mu_k)),$$

where $\tau \in (0,\infty)$ is Gamma(c,d) and $\mu \in$Simplex of R^k is Dirichlet($m_1,\ldots, m_k$).

Example 11: Note that

$$V(\theta_{ij}|\tau,\mu) = \mu_j(1-\mu_j)/(1+\tau)$$

Hence, for fixed μ the larger τ the smaller Var(θ_{ij}|τ,μ)

When τ tends to infinity the variance of θ_{ij} vanishes, and the limiting model is the non-hierarchical multinomial.

By estimating τ from the data, we determine how much heterogeneity exists across precincts, and learn whether the hierarchical model is called for.

Example 11: When we care about the distribution of θ_i, the statistical model for $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ can be posed as:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|\tau,(\mu_1,\ldots,\mu_k) \sim \prod_{i=1}^{n} \text{DirichletMultinomial}(n_i,\tau(\mu_1,\ldots,\mu_k)),$$

and the observations are i.i.d. DirichletMultinomial(n_i,τμ), with prior ψ(τ,μ).

Example 11: When observations are i.i.d. DirichletMultin(n_i,τμ),

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|\tau,(\mu_1,\ldots,\mu_k) \sim \prod_{i=1}^{n} \text{DirichletMultinomial}(n_i,\tau(\mu_1,\ldots,\mu_k)),$$

we have that:

$$E(\tilde{y}_{ij}|\tau,\mu) = n_i\mu_j,$$

$$V(\tilde{y}_{ij}|\tau,\mu) = n_i\mu_j(1-\mu_j)\frac{n_i+\tau}{1+\tau},$$

$$Cov(\tilde{y}_{ij},\tilde{y}_{ih}|\tau,\mu) = -n_i\mu_j\mu_h\frac{n_i+\tau}{1+\tau}.$$

Different from the independent Multinomial(θ) model, this allows us to model variance and covariance separate from the mean.

Example 11: When we care about the distribution of $\tilde{y}_i$, the statistical model for $\tilde{y}=(\tilde{y}_1,\ldots,\tilde{y}_n)$ can be posed as:

$$(\tilde{y}_1,\ldots,\tilde{y}_n)|(\theta_1,\ldots,\theta_n) \sim \prod_{i=1}^{n} \text{Multinomial}(n_i,\theta_i),$$

where the prior for $\theta = (\theta_1,\ldots,\theta_n)$ is:

$$(\theta_1,\ldots,\theta_n) \sim \int \prod_{i=1}^{n} \pi(\theta_i|\tau,\mu)\psi(\tau,\mu)d\tau d\mu,$$

where $\pi(\theta_i|\tau,\mu)$ is $\text{Dirichlet}(\tau(\mu_1,\ldots,\mu_k))$.

Example 11:

```
library(R2jags)

model.multhier1 <- "
model
{
  for (i in 1:nn) {
    y1[i,1:k] ~ dmulti(pi1[i,1:k],Vot[i])
    pi1[i,1:k] ~ ddirch(alf[1:k])
  }
  tau ~ dgamma(3,0.01)
  a = rep(1,k)
  mu ~ ddirch(a)
  for (j in 1:k) {
    alf[j]=tau*mu[j]
  }
  }
"

  data <- list(nn=dim(y)[1], k=dim(y)[2], y1=y, Vot=P06$elec)

#  parameters <- c("pi","tau","mu","alf")
  parameters <- c("tau","mu","alf")


  example1 <-jags(data, parameters.to.save=parameters,
               model=textConnection(model.multhier1), n.iter=3000, n.burnin=1000, n.thin=2,n.chains=3)

  print(example1)
```